

Sukesh Periyasamy

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RESEARCH PROFILE

Early-career researcher with Master in Medical Technology from IIT Jodhpur (joint programme with AIIMS Jodhpur) and currently working as Junior Research Fellow. Interdisciplinary experience spans biomedical sensing, cleanroom microfabrication, spectroscopy, healthcare technology, and data-driven research. Research focus: integrating computational analysis, biomedical instrumentation, and machine learning to support clinical decision-making. Skilled in SERS substrate development, signal processing (EMG), scientific data pipelines, experimental characterisation, and prototype calibration. Committed to developing interpretable, clinically grounded AI systems for early disease detection, specifically neonatal sepsis, and personalized medicine.

EDUCATION

- Indian Institute of Technology Jodhpur (Joint with AIIMS)** Rajasthan, India
Master in Medical Technology; CGPA: 8.53 / 10 2024 - 2026
- Erode Sengunthar Engineering College** Tamil Nadu, India
Master of Computer Applications; CGPA: 7.20 / 10 2022 - 2024
- Kongu Arts and Science College** Tamil Nadu, India
B.Sc. Information Technology; CGPA: 6.90 / 10 2019 - 2022

TECHNICAL SKILLS

- Programming:** Python, JavaScript
- Scientific Computing:** NumPy, Pandas, Matplotlib, Jupyter Notebooks
- Data Analysis:** Statistical analysis, signal processing, data preprocessing
- Machine Learning:** Feature engineering, supervised learning, model evaluation
- Signal Processing:** EMG acquisition, filtering, envelope detection, calibration
- Spectral Analysis:** Raman spectroscopy, FTIR, SERS data processing, DFT simulations
- Fabrication:** Photolithography, PVD (Sputtering, Thermal & E-beam Evaporation), SEM
- Web & Mobile:** Android SDK, FastAPI, React, Tailwind CSS, Redis
- Software & Platforms:** Git, GitHub, Linux, Unity 3D, Arduino, Raspberry Pi

RESEARCH EXPERIENCE

- Indian Institute of Technology Jodhpur** Rajasthan, India
Junior Research Fellow (JRF) - Funded by TCS 2026 - Present
 - Developing silicon-based Surface-Enhanced Raman Spectroscopy (SERS) biosensors for sensitive, label-free biomarker detection targeting neonatal sepsis.
 - Conducting cleanroom microfabrication processes including photolithography and Physical Vapor Deposition (PVD) systems (sputtering, thermal evaporation, e-beam evaporation) to design and fabricate nanostructured sensing platforms.
 - Performing structural and morphological characterisation of fabricated substrates using Scanning Electron Microscopy (SEM) and analysing micro/nanostructures.
 - Developing Python-based computational pipelines for Raman, FTIR, and SERS spectral preprocessing, including ALS baseline correction, Savitzky-Golay smoothing, peak detection, and spectral feature extraction.
 - Validating molecular spectra using Density Functional Theory (DFT) simulations (Gaussian 16, Avogadro) against experimental Raman and FTIR measurements to accurately identify biomarkers such as LPS, LTA, and NAM.
 - Conducting experimental optimisation of nanostructured sensing platforms to improve spectral enhancement, reproducibility, and sensitivity for biomarker applications.

MANUSCRIPTS IN PREPARATION

- Experimental and DFT-Assisted Raman Characterisation of N-Acetylmuramic Acid** IIT Jodhpur 2026
Manuscript in Preparation – Bacterial Biomarker Applications
 - Acquired high-quality Raman spectra of bulk N-acetylmuramic acid (NAM) under optimized conditions and developed a complete data-processing workflow (baseline correction, smoothing, normalization, reproducibility analysis) to establish a validated reference spectrum for bacterial cell-wall studies.
 - Performed Density Functional Theory (DFT) calculations to simulate Raman spectra and accurately assign theoretical vibrational modes, demonstrating the combined utility of experimental and computational methods for identifying NAM in Raman-based diagnostic applications.

PROJECTS

- Silicon-Based SERS Biosensor Development (Ongoing):** Designing and fabricating silicon-based SERS substrates in a cleanroom environment via photolithography and PVD (sputtering, thermal/e-beam evaporation) for improved plasmonic enhancement. Characterising nanostructured substrates using SEM to evaluate morphology. Developing Python-based computational workflows for Raman, FTIR, and SERS spectral preprocessing for neonatal sepsis biomarker identification (Funded by TCS). Tech: Python, DFT, Raman, FTIR, SEM.
- AnteClick – Real-Time Mobile Banking Phishing Protection Platform:** Designed and developed an Android cybersecurity platform and SDK to detect phishing URLs and fake banking apps in under 300ms. Built an event-driven engine using Android Accessibility Service for heuristic analysis (typosquatting, homographs) without inspecting private data. Developed a scalable FastAPI backend with Redis caching, integrating threat intelligence feeds. Tech: Jetpack Compose, FastAPI, Redis, Python, React.
- NeuroSync – EMG-Based VR Game Controller:** Built a real-time EMG-based control system for a Unity VR football game. Implemented full signal pipeline: amplification, filtering, envelope detection, and peak confirmation with auto-calibration and rise-only triggering. Mapped muscle activations to game actions via serial communication. Tech: Arduino Uno, BioAmp EXG Pill, Unity 3D.

PROFESSIONAL EXPERIENCE

- **Jodhpur City Knowledge and Innovation Foundation (JCKIF)** Jodhpur, India
Internship - Product Management Oct 2024 - Mar 2025
 - Managed digital product catalogue ensuring accurate listings and metadata consistency.
 - Captured and edited product images to enhance visual quality and user appeal.
- **Atos Syntel** India
Internship - Software Testing Oct 2022 - Nov 2022
 - Performed manual and automation testing (functional, integration, regression) within an Agile development environment, collaborating with teams on bug tracking.
- **KEC College** Tamil Nadu, India
Workshop Instructor – 4-Day Technical Training 2024
 - Conducted a hands-on technical workshop for 35+ students covering Computer Hardware, Linux, Web Development, Git, Cloud, and IoT fundamentals.

HONORS AND AWARDS

- Finalist – Smart India Hackathon 2024
- 1st Prize – Website Creation Competition
- 2nd Prize – Code Fighter Competition
- Bronze Medal – Inter-Collegiate Taekwondo
- Represented Tamil Nadu – Senior National Softball Championship